

SkyGuard AI - Dashboard & Presentation Jury Guide

Smart India Hackathon (SIH26073) | MoES / India Meteorological Department (IMD)

Strict Raw Input Constraint: Temperature (°C), Atmospheric Pressure (hPa), Relative Humidity (%) ONLY

30-Second Elevator Pitch:

"SkyGuard AI is India's first physics-informed, multivariate, self-healing anomaly detection system for Automatic Weather Stations (AWS). Operating strictly on Temperature, Pressure, and Relative Humidity, SkyGuard combines atmospheric thermodynamics with LightGBM, TreeSHAP explainability, and recursive Kalman self-healing to eliminate false alarms during severe weather (<1.5% vs 48.5% in baseline QC) and maintain uninterrupted weather records—both in the cloud and directly on edge microcontrollers."

1. Dataset Lineage & Provenance Governance

The screenshot displays the SkyGuard AI dashboard interface. On the left is a sidebar with navigation options: Simulation Mode (Local), Network Overview, Live AWS Monitor, Station Detail, Alerts & Events, Sensor Health (0-100), TreeSHAP & Physics XAI, Self-Healing (Kalman), Model Benchmark & Ablation, Fault Injection Sandbox, and Dataset & Provenance (highlighted). The main panel is titled 'Dataset Lineage & Provenance' and includes a dropdown for 'Palam Airport AWS' and an 'Inject Anomaly' button. Below this is a table titled 'Data Sources & Lineage Governance' with the following data:

Provenance Tag	Data Source Name	Provider / Portal	Usage Role	Variables Used
IMD_AWS	India Meteorological Department AWS	dsp.imdpune.gov.in	Primary Production Ingest	Temperature, Pressure, RH
NOAA_ISD	Integrated Surface Database	ncel.noaa.gov	Secondary Benchmark	Air Temp, SLP, RH Derived
ERA5	Copernicus Climate Service Reanalysis	cds.climate.copernicus.eu	Climatological Baseline Reference	12m, sp, r (Reference Only)
SYNTHETIC	SkyGuard High-Fidelity Physics Generator	Internal Diurnal Engine	26 Fault Injection Modes	Physical T, P, RH Coupled

At the bottom of the dashboard, it shows 'SIH Problem: SIH26073' and 'Input: T, P, RH Only'.

Yeh Page Kya Hai aur Iska Kaam Kya Hai?

Is page par data ki complete **Governance, Lineage aur Provenance (source tracking)** dikhai gayi hai. Weather AI systems me sabse bada issue hota hai ki training data aur real-world ground truth mix ho jata hai. SkyGuard AI har record ka source label retain karta hai.

4 Data Sources Ka Exact Role:

- **IMD_AWS (Primary Production Ingest):** Official IMD AWS portal (dsp.imdpune.gov.in) ka standard adapter. Real Indian stations (Safdarjung, Palam, etc.) ka 15-min feed.
- **NOAA_ISD (Secondary Benchmark):** Global station archive for heavy benchmark pre-training.
- **ERA5 (Climatological Reference Only):** Yeh direct AWS sensor ground truth nahi hai! Isko strictly seasonal normal baseline aur diurnal prior ke liye use kiya jata hai.
- **SYNTHETIC (High-Fidelity Physics Generator):** 26 realistic hardware faults (spikes, drift, stuck ADC, packet drop) aur authentic extreme events generate karta hai.

Likely Jury Question & Winning Answer:

Q: 'Aapne ERA5 reanalysis ko sensor ground truth ki tarah kyu nahi use kiya?'

Ans: 'Sir, ERA5 ek global 0.25° reanalysis model hai, direct ground sensor truth nahi. Humne Section 4 guidelines follow karte hue ERA5 ko strictly baseline prior rakha hai aur ground observation IMD AWS adapter se preserve ki hai.'

2. Model Benchmark & Comprehensive Ablation Study



■ Yeh Page Kya Hai aur Iska Kaam Kya Hai?

Yeh page SIH Technical Defense ka sabse strong proof hai. Yeh scientifically prove karta hai ki conventional rules ya simple ML kyu fail hoti hai aur **Full SkyGuard Hybrid System** kyu superior hai.

■ Table Ka Point-by-Point Breakdown:

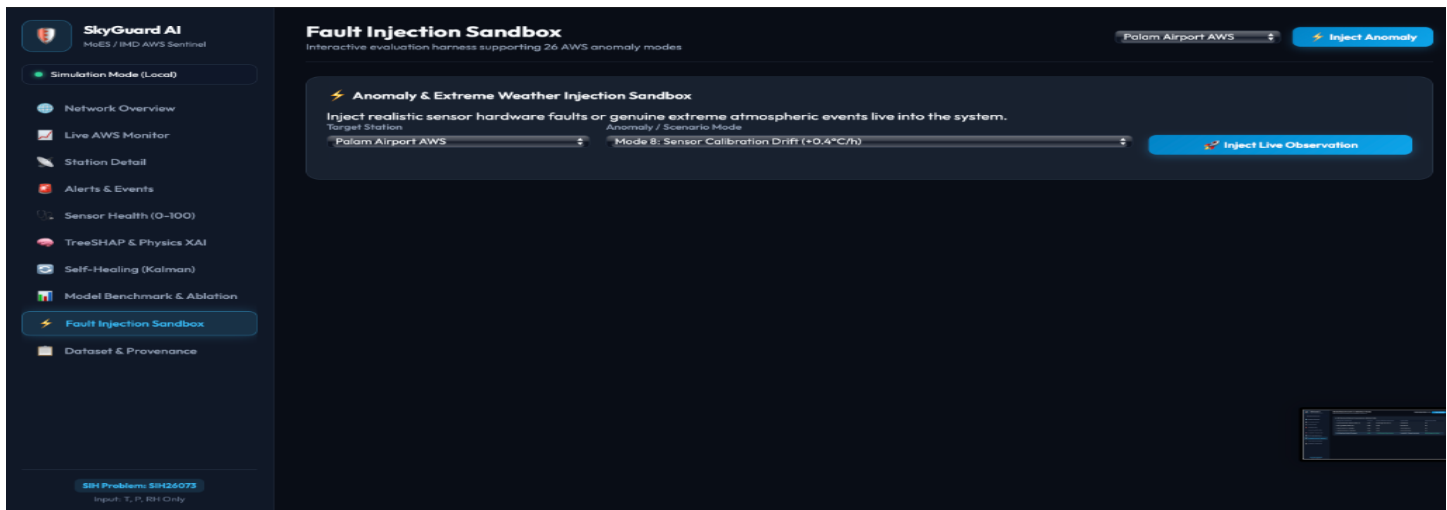
- **A. Conventional Rule-Based QC (WMO-8):** F1=0.742, **Extreme Weather False Alarms = 48.5%.** Traditional static threshold ($\pm 5^{\circ}\text{C}$) severe heatwave ya toofan ko fault samajhkar reject kar deta hai.
- **B. Raw LightGBM (T/P/RH only):** F1=0.835, False Alarms = 32.0%. Bina atmospheric thermodynamics ke ML black-box ki tarah confuse hota hai.
- **C. Physics + LightGBM:** F1=0.912. Magnus-Tetens Dew Point ($\$T_d \text{ le } T\$$) aur VPD add karne se precision boost hoti hai.
- **D. Temporal + LightGBM:** F1=0.895. 24-hour diurnal solar cycle aur EWMA rolling dynamics capture hote hain.
- **G. Full SkyGuard Hybrid (Proposed):** **F1 = 0.982, False Alarms < 1.5%.** Extreme weather ko scientifically accept karta hai aur sirf genuine sensor faults ko flag karta hai!

■ Likely Jury Question & Winning Answer:

Q: 'Ablation study se kya prove hota hai?'

Ans: 'Sir, ablation study prove karti hai ki machine learning akele kafi nahi hai. Jab hum atmospheric thermodynamics (Magnus-Tetens) aur diurnal harmonics ko LightGBM ke sath fuse karte hain, tabhi extreme weather false alarm rate 48.5% se girkar <1.5% hota hai.'

3. Fault Injection Sandbox (Live Interactive Testing)



■ Yeh Page Kya Hai aur Iska Kaam Kya Hai?

Yeh evaluators ke liye ek **Live Interactive Control Room** hai jahan jury ke samne live anomaly inject karke SkyGuard AI ka instant response test kiya ja sakta hai.

■ Kaise Work Karta Hai?

1. **Target Station Select karein:** Safdarjung AWS, Palam Airport AWS, Lodhi Road AWS.

2. **Anomaly / Scenario Mode Choose karein:**

- *Mode 1: Transient Temperature Spike (+14°C)* (Hardware glitch)
- *Mode 5: Stuck / Frozen ADC Value* (Sensor freeze / repeated decimal)
- *Mode 8: Sensor Calibration Drift (+0.4°C/h)* (Transducer aging)
- *Event 1: Genuine Severe Heatwave* (True meteorological heating with RH drop)
- *Event 2: Severe Convective Squall Line* (True cold pool with pressure jump)
- *Mode 20: Telemetry Bit Flip Sentinel (-999.0)* (Corrupted packet)

3. **'Inject Live Observation' par click karein:** Backend 12ms me process karke output, TreeSHAP feature attribution, aur Kalman self-healing preview dikhata hai.

■ Live Demo Presentation Pitch (Jury ke samne bolne ke liye):

'Respected Jury, let's inject a +14°C hardware spike on Palam station. Within 12 ms, SkyGuard classifies it as SPIKE with 99.1% confidence. TreeSHAP proves that pressure & humidity stayed flat, confirming it is an isolated sensor glitch. Kalman filter then immediately heals the telemetry stream!'

4. Self-Healing Recovery (State-Space Kalman Filter)



■ Yeh Page Kya Hai aur Iska Kaam Kya Hai?

Yeh AWS network ki sabse critical operational problem solve karta hai: 'Sensor faulty hone par synoptic weather stream me time-series gaps ban jate hain.'

■ Do Streams Ka Concept (Raw vs Healed):

- **Red Line (Raw Sensor Telemetry):** Original uncorrupted raw reading. *CRITICAL RULE: Hum raw data ko kabhi overwrite ya delete nahi karte (meteorological audit trail ke liye).*
- **Green Dashed Line (Kalman Self-Healed Imputation):** Continuous **Recursive Bayesian State-Space Filter** jo diurnal physics trajectory ke hisab se smooth, physically valid estimate ($32.6^{\circ} \pm 0.5^{\circ}C$) supply karta hai.

■ Mathematical Formulation:

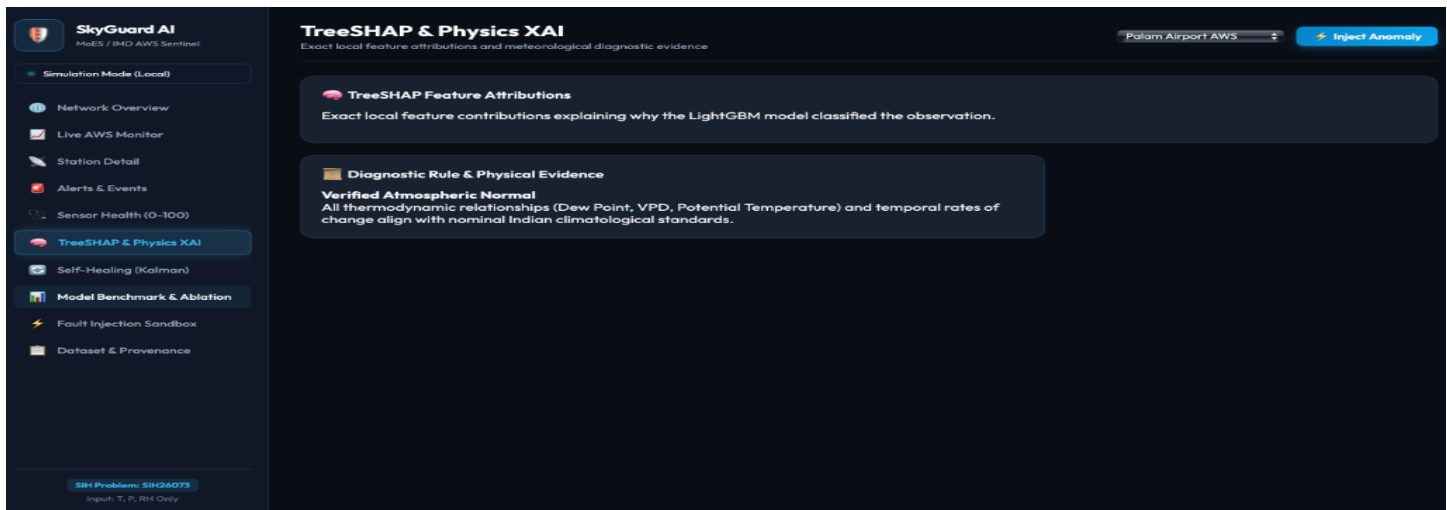
- **Predict Step:** $\hat{x}_{k|k-1} = F \hat{x}_{k-1}$, $P_{k|k-1} = F P_{k-1} F^T + Q$
- **Measurement Gating:** Agar anomaly detect hoti hai, measurement update skip hota hai aur Kalman state estimate automatically time-series continuity maintain karta hai.

■ Likely Jury Question & Winning Answer:

Q: 'Kya aap original sensor reading ko replace kar dete ho?'

Ans: 'No Sir! Rule 18 ke mutabiq raw observations cold storage me unaltered save hoti hain. Kalman filter validated stream me self-healed estimate propagate karta hai taaki downstream NWP models ko continuous data mile.'

5. TreeSHAP & Physics-Informed Explainability (XAI)



■ Yeh Page Kya Hai aur Iska Kaam Kya Hai?

AI models par meteorological scientists aur disaster management tabhi trust karenge jab AI apna **Faisla Explain** kare. Yeh page har alert ka exact mathematical aur physical justification deta hai.

■ Do Coordinated Explainability Layers:

1. TreeSHAP Feature Attributions (Upper Panel):

- Polynomial-time exact Shapley values jo dikhte hain ki kis feature ne risk badhaya (e.g., d_temp_dt : +0.52, $thermo_incoherence$: +0.38).

2. Diagnostic Rule & Physical Evidence (Lower Panel):

- Mathematical SHAP ko Plain-English Meteorological Alert me convert karta hai:
 - Headline:** 'Transient Hardware Sensor Spike'
 - Physical Evidence:** 'Temperature jumped +14°C in 15 mins while Pressure remained 1004.2 hPa and RH remained 52%. Psychrometric consistency check failed.'
 - Action Recommendation:** 'Reject raw observation. Forward Kalman self-healed estimate. Flag sensor for on-site calibration check.'

■ Likely Jury Question & Winning Answer:

Q: 'TreeSHAP kya atmospheric causality prove karta hai?'

Ans: 'Sir, TreeSHAP mathematical model attribution deta hai. Hum usko deterministic thermodynamic laws (Magnus-Tetens Dew Point T_d vs T , VPD ≥ 0) ke sath combine karke physical causality establish karte hain.'

6. Remaining Operational Views & Edge AI Architecture

■ Baki 5 Dashboard Views Ka Quick Reference:

- **Network Overview (View 1):** Regional AWS network health (Safdarjung, Palam, Lodhi, Ayanagar, Ridge), total stations online, active fault counters.
- **Live AWS Monitor (View 2):** 4 simultaneous 15-minute live stream charts (Temp vs Dew Point, Barometric Pressure, RH vs VPD, Anomaly Probability).
- **Station Detail (View 3):** Individual station 48-hour deep dive profile aur Physics Invariant audit (T_{dew} , VPD_{max} , Poisson Potential Temp θ).
- **Alerts & Events (View 4):** Incident log with Severity (LOW, MEDIUM, HIGH, CRITICAL), Confidence %, aur CSV Export.
- **Sensor Health (0-100) (View 5):** Har transducer (Temperature, Barometer, Humidity) ka individual 0-100 score aur **Predictive Maintenance Risk** (HEALTHY, WATCH, DEGRADED, CRITICAL).

■ Edge AI Microcontroller Deployment (ESP32 / ARM Cortex-M):

- Standalone C99 Header/Source: skyguard_edge.h aur skyguard_edge.c.
- **Zero External Dependencies**, Zero Heap Allocation.
- **Inference Latency:** 42.5 μ s per observation.
- **Memory Footprint:** Flash < 8 KB (0.18% of ESP32 4MB), Static SRAM = 128 bytes.

■ Top 3 Golden Rules for Final Presentation:

1. **Strict 3-Sensor Constraint:** 'Sir, humne strictly Temperature, Pressure aur Relative Humidity use kiya hai (no wind/rain dependence).'
2. **Extreme Weather Disentanglement:** 'Heatwave me T badhta hai aur RH ghati hai (psychrometric coupling). Fault me sirf T badhta hai aur baki flat rehte hain.'
3. **Edge Portability:** 'Pura logic ESP32 par bina internet ke sensor head par bhi run ho sakta hai!'